

Oliver Paynter-Jones

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EDUCATION

University of Bath

Bath, UK

BSc(Hons) Computer Science and Mathematics, **First Class Honours**

Oct. 2022 – Jul. 2025

Honours Thesis: Behavioural Biometrics: ML-Based Anomaly Detection for Anti-cheat (**75%**)

[View Full Thesis](#)

Christ the King College

Isle of Wight, UK

A-Levels: Further Mathematics, Mathematics, Computer Science, EPQ (**A***); Economics (**B**)

Sep. 2020 – Jul. 2022

EXPERIENCE

Software Development Engineer Intern

Jun. 2025 – Aug. 2025

Amazon

Edinburgh Development Centre, UK

- Took ownership of an open-ended problem, working closely with non-technical stakeholders to translate operational pain points into a tool that automated product intake and delivery tracking.
- Conducted interviews and gathered requirements; authored comprehensive documents outlining system designs and recommending a chosen solution.
- Led reviews with peers and senior engineers to gather feedback and secure alignment on the selected approach.
- Designed and implemented a TypeScript-based browser tool integrating internal issue management APIs with external software, replacing an error-prone manual workflow with a one-click or automated background process.
- Applied internal design standards and simplified setup and configuration, cutting onboarding time and making the tool accessible to a wider range of users.
- Integrated with internal build and deployment systems, implementing unit tests and CI/CD pipelines to ensure maintainability, reliability, and adherence to engineering best practices.
- Produced technical documentation and delivered a final presentation and live demo to engineers and managers, receiving highly positive feedback on the solution's quality, usability, and alignment with business needs.
- Received positive feedback for taking initiative, driving outcomes, fostering strong trust with stakeholders and team members, and performing deep analytical work to inform decisions.

PROJECTS

The Perfect Keyboard: A Genetic Algorithm Experiment | Python

[Blog](#)

- Built a genetic algorithm to optimise keyboard layouts for typing speed and comfort using frequency data from a 246K-word corpus.
- Implemented custom crossover and mutation operators to evolve and identify the most efficient layout.

Society Matchmaker | React, Flask, SQL, Git

[Blog](#)

- Led an 8-member team tackling low student-society engagement; coordinated design, CI/CD, and version control practices.
- Developed key frontend features including user management, interest polling, and personalised event recommendations.
- Achieved a 150% improvement in student-society engagement across two key metrics compared to existing systems.

Solving the Rubik's Cube with Group Theory | Python

- Researched and applied advanced mathematical group theory to model cube states and transitions efficiently.
- Engineered a Python solver using NumPy and pandas with custom cube encoding and decoding algorithms.
- Implemented a modified IDA* search with pruning, achieving sub-second solve times for any scramble.

SKILLS

Programming: Typescript, Java, Python, C++, Javascript, SQL

Infrastructure: AWS, CI/CD, Docker, Version Control (Git)

Frameworks/Libraries: React, Node.js, Flask, Webpack, JUnit, Jest, GitHub